PORTFOLIO CASE STUDY · NOVEMBER 2026

India Semiconductor Industry

Market Opportunity and Strategic Investment Analysis, 2026–2030

Case studies: Micron Technology (Sanand) and Tata Electronics (Dholera and Assam)

THE ANSWER

Partly. By 2030 India can be a credible chip-packaging location with about 1–2% of world capacity, cost-competitive only because of subsidy. A mature-node fab is possible but loses money until the early 2030s; leading-edge fabrication is not realistic.

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Source: India Semiconductor Industry report, Nov 2026 (Executive summary). ₹ converted at ₹95 per US$ unless a company disclosed US$.

APPROACH

I tested the packaging-first hypothesis against three measurable tests of “global competitiveness”

> **STARTING HYPOTHESIS (TESTED, NOT ASSUMED)**
>
> *By 2030 India can compete in assembly, test and packaging and in selected mature-node and compound-semiconductor manufacturing; leading-edge fabrication will remain difficult.*

TEST A

Domestic manufacturing contribution

Share of India’s chip demand made, packaged and value-added in India

TEST B

Position in the global supply chain

India’s share of global packaging and fab capacity

TEST C

Manufacturing economics

Plant cost and return vs Malaysia and Vietnam, with and without subsidy

SQL

SQLite tracker of the 12 approved units and their concentration

Python

Five pandas and numpy notebooks; every CSV carries a source column

Monte Carlo

10,000 runs, triangular inputs, ρ = 0 and 0.7, for Sanand and Dholera

Excel model

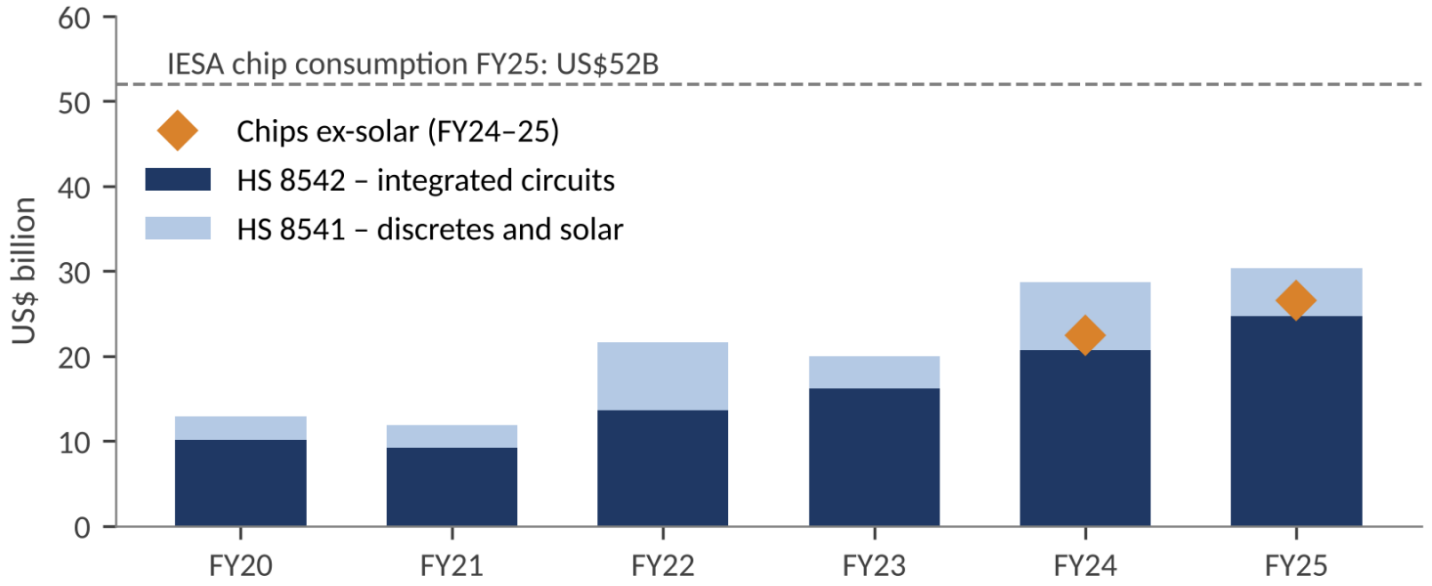
One workbook of live formulas with scenario and support selectors

Source: report Section 1, Annex A and Annex G. Data: PIB, ISM, MeitY, PRS, TradeStat; Micron, Tata Sons, UMC, PSMC, Amkor filings; WSTS, SIA–BCG, SEMI, IESA, JETRO, TrendForce.

INDIA MARKET

India uses about US$52 billion of chips a year but makes almost none; even in 2030 only 1.3% of the value is added here

Loose-chip imports vs chip consumption, US$ billion



US$52B

Chips used in India in FY25 (IESA), 6–8% of world demand; ~US$96B by 2030 in the base case.

US$26.5B

Loose-chip imports ex-solar in FY25: about half the market. The rest arrives inside imported boards.

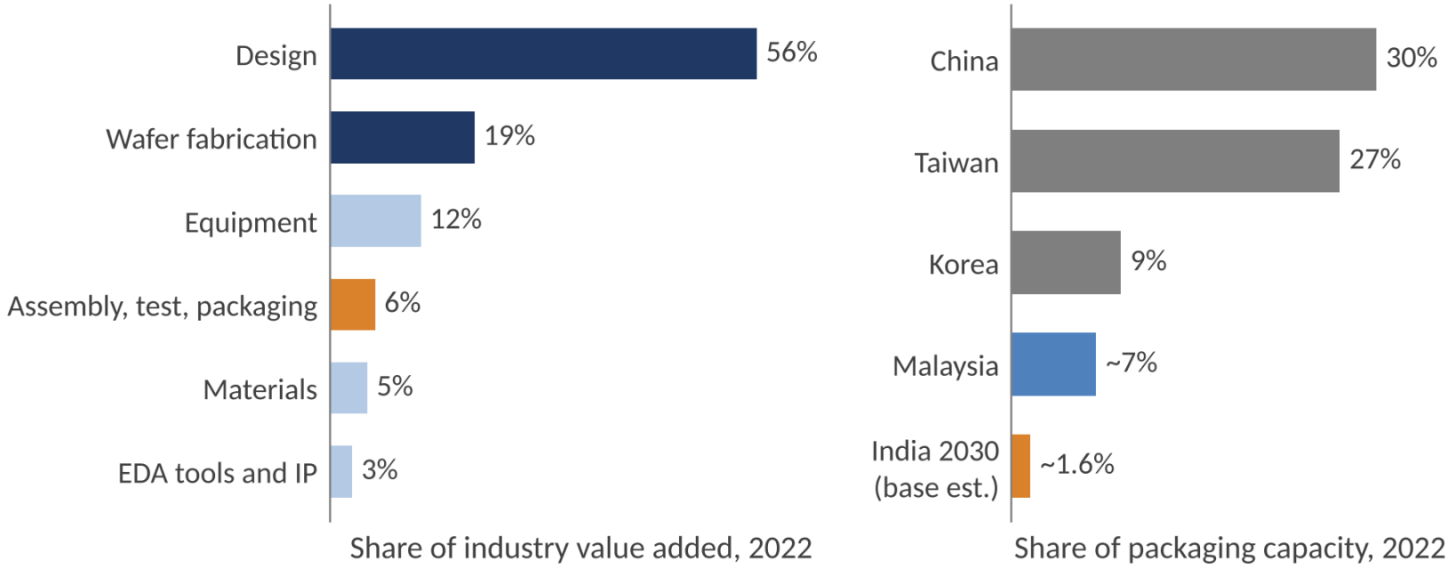
1.3%

Of 2030 demand value-added in India (base). 7.1% is made or packaged here; 1.7% reaches Indian buyers.

Source: TradeStat, HS 8541 and 8542 (Ref. 11); IESA, Feb 2025 (Ref. 28); solar split via busy.in; report Sections 3 and 9.1 (base case).

Design and fabs hold three-quarters of the industry's value; packaging holds only 6% but is the easiest way in

Value added by activity (left) and packaging capacity by location (right)



75%

Of industry value sits in design (56%) and wafer fabrication (19%). Packaging is 6%.

17%

Typical OSAT gross margin, 2016–19. Thin, but packaging needs far less capital and know-how than a fab.

~1.6%

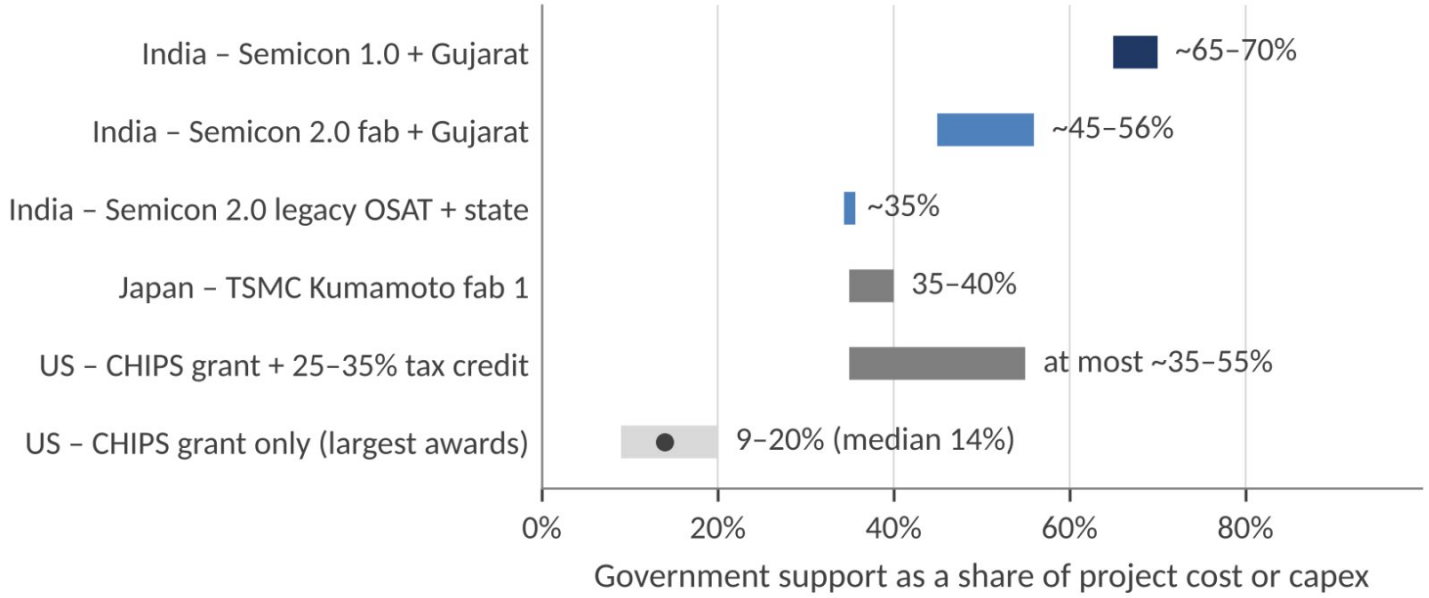
India's base-case share of global packaging in 2030 (own estimate, by value) vs Malaysia's ~7% today.

Source: SIA–BCG (2024), Exhibits 2, 6 and 13 (Ref. 24); SIA–BCG (2021) (Ref. 25); India 2030 is an own estimate (Section 5.4, Annex A.6).

POLICY

Semicon 1.0 paid ~65–70% of project cost, more than any country compared; Semicon 2.0 cuts manufacturing support

Government support as a share of project cost or capex



₹1.64 lakh crore

12 projects approved under Semicon 1.0 (₹76,000 crore outlay); three packaging plants producing in 2026.

25–40%

Semicon 2.0 central support (July 2026), on eligible capex that excludes land, IDC and technology transfer.

~7%

Of the Semicon 1.0 outlay actually spent by 2025-26: money is released only as plants are built.

Source: ISM guidelines (Refs. 2–4); PIB and gazette (Ref. 1); Gujarat policy (Ref. 12); GAO-26-107882 (Ref. 13); report Exhibit 4, Sections 4.2, 4.4, 5.2.

BENCHMARKING

Against Malaysia and Vietnam, India wins on subsidy and home market; labour cost barely moves the economics

How India compares with its real packaging competitors

Factor	India vs Malaysia	India vs Vietnam
Labour (median factory worker, JETRO FY2023)	Cheaper (US$294 vs 403)	Dearer (US$294 vs 251)
Capex support	Much higher	Much higher
Supplier base, logistics	Weaker	Similar or weaker
Existing customers	Few	Fewer
Domestic chip demand	Much larger (US$52B)	Larger

India's real gaps are a missing supplier base and no track record with global customers: every running merchant unit has a foreign chip company as its first customer.

±2–3%

Of an OSAT's sales: what India's labour gap is worth, as labour is ~10% of sales (Amkor).

~18 pts

Of revenue: how far the capex subsidy can move depreciation, about six times the labour effect.

~1–1.6%

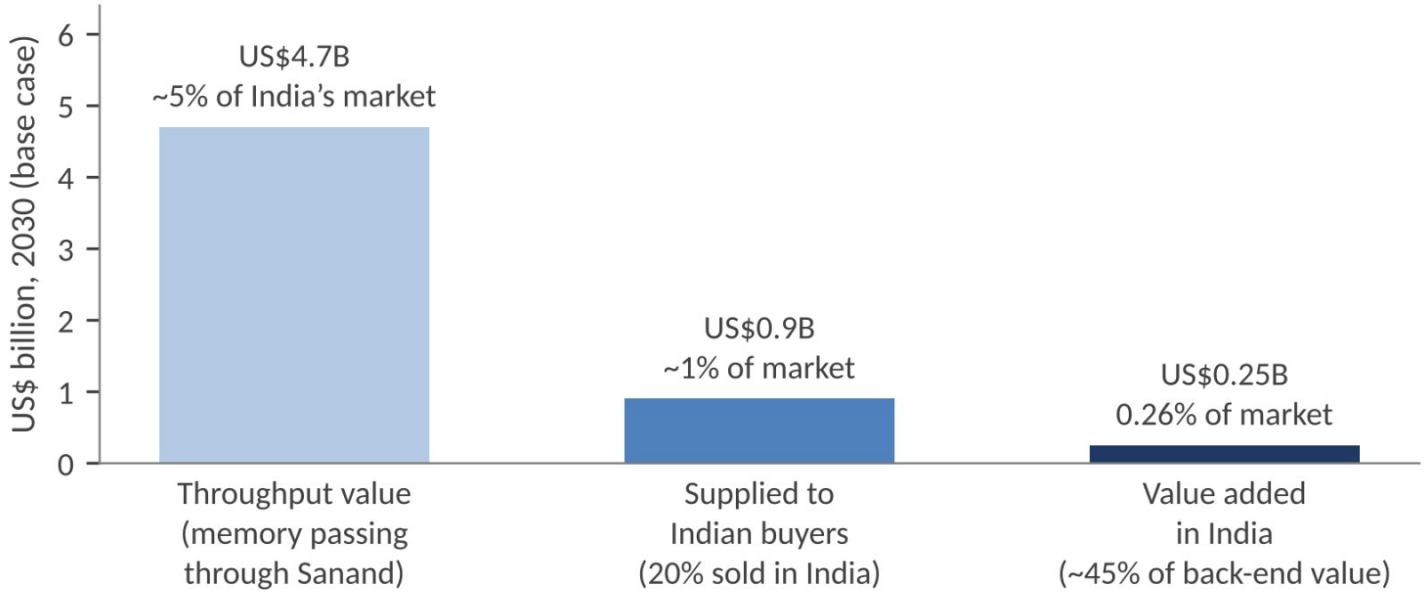
India's share of global packaging by 2030, about a fifth of Malaysia's ~7% today.

Source: JETRO FY2023 (Ref. 30); Amkor 10-K 2025 (Ref. 19); MITI National Semiconductor Strategy (Ref. 14); report Exhibit 6, Sections 4.3, 5.3, 5.4, 6.1.

MICRON SANAND

Micron Sanand moves ~US$4.7 billion of memory in 2030, but only ~0.26% of India's chip market is value added here

Sanand in 2030, base case: what passes through vs what stays in India



<3 years

From approval (Jun 2023) to commercial production and first modules to Dell (Feb 2026).

US$4.7B

Base case: 55% of 1,352M units = 744M units at US$6.26. Monte Carlo band US$2.9–8.7B (ρ = 0.7).

US$825M

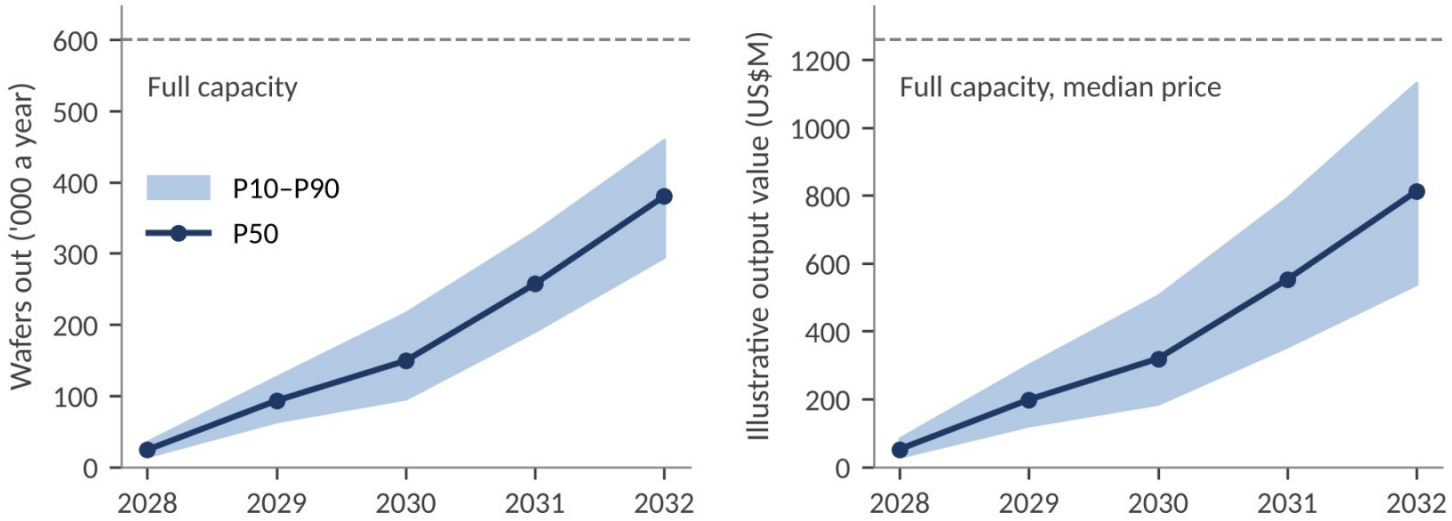
Micron's share of the US$2.75B cost; government paid 70%. Wafers come in and most output goes out.

Source: PIB, 28 Feb 2026 (Ref. 8); PRS; Micron FY2026 10-K (Ref. 15); TrendForce (Ref. 31); own scenarios, Sections 7.3–7.4. Throughput value, not Sanand revenue.

TATA ELECTRONICS

Tata Dholera's constraint is customers, not cost: in the middle case it runs at about a quarter of capacity in 2030

Dholera ramp, P10–P90 band and P50: wafers out (left) and illustrative output value (right)



~25%

Of design capacity in 2030 (middle case): ~149,000 wafers, US$0.32B vs ~US$1.26B at full load.

46–88%

Peer break-even loading: old fabs ~46%, new fabs 80–88%. PSMC turned positive only at 82%.

2–3 years

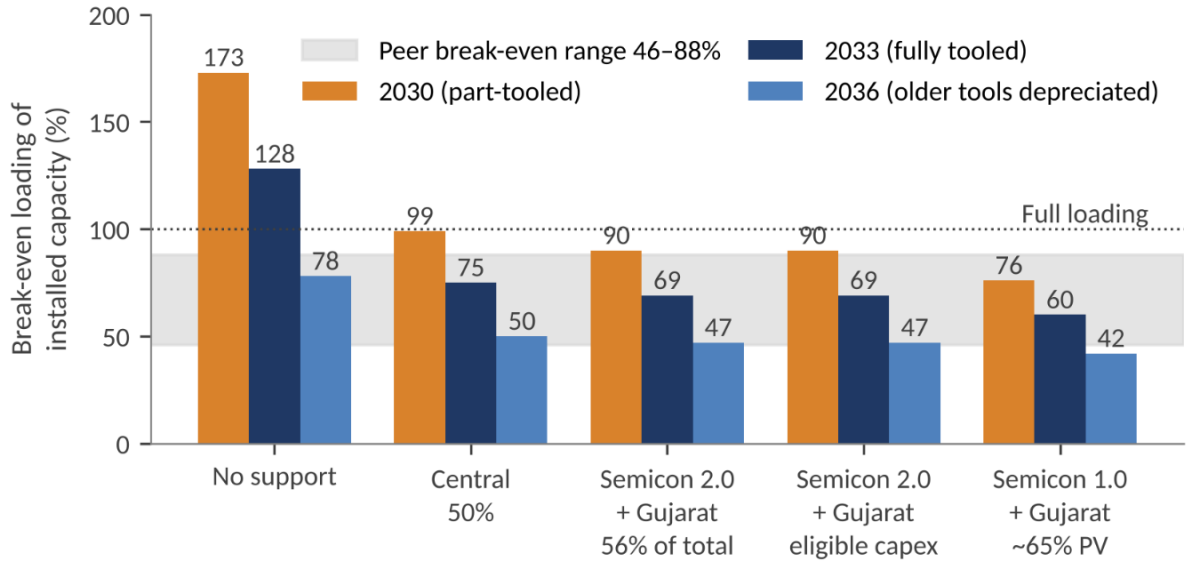
For automotive qualification. Most chips used in India are chosen by global firms.

Source: own Monte Carlo (phase6_tata.ipynb); peer filings 2021–25 (Refs. 17, 18, 21); PIB (Ref. 5); Sections 8.3–8.4. Illustrative output value, not a Tata revenue forecast.

FINANCIAL ANALYSIS

Subsidy turns an unviable fab or packaging plant into a workable one, but a new fab still earns below a normal hurdle rate

Reference fab: break-even loading of installed capacity (%)



Reference packaging plant: 10-year IRR

Support case	IRR	Payback
No support	−1.1%	None in 10 yrs
Semicon 2.0 legacy (35%)	5.4%	Year 8
Semicon 2.0 advanced (49%)	9.3%	Year 7
Semicon 1.0 central (50%)	9.7%	Year 7
Semicon 1.0 + Gujarat (~65% PV)	15.9%	Year 6

Fab 20-year IRR: at best 7.0% with ~65% support and 1.8% on Semicon 2.0 eligible-capex rules, below a 10–12% hurdle. **Price matters most:** ±10% moves the packaging IRR from 0.7% to 27.6%.

Source: workbook, Financial scenarios tab, blocks C and D; phase7_models.ipynb. Illustrative reference facilities (Amkor- and UMC-based), not company economics.

RECOMMENDATIONS

India should bet on packaging now and design IP later, finish Dholera before any second fab, and track customers, not floor space

Recommendation	Why (evidence)	Metric to watch
1. Make packaging the core bet; pay for customers, not only capex	Utilisation and price move a plant's IRR far more than labour or yield	Utilisation of merchant packaging plants
2. Finish Dholera; do not fund a second fab on Semicon 2.0 terms	On eligible-capex rules a new fab's 20-year IRR is about 1.8%	Dholera's loading in 2030–32
3. Turn design talent into Indian-owned chip IP	~20% of the world's chip designers, but Indian design firms earn under ₹150 crore	Revenue of Indian-owned design companies
4. Build the supplier base inside the Gujarat cluster	About 55% of a packaging plant's cost is materials, almost all imported	Local share of materials in Indian packaging plants
5. Publish milestone-based payments so money flows as plants are built	Only ~7% of the Semicon 1.0 outlay was spent by 2025-26	Central disbursement as a share of committed support

Source: report Executive summary and Sections 9–10 (recommendations 10.1–10.8); workbook, Financial scenarios tab.